

Akshay Bapat

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EDUCATION

Georgia Institute of Technology, Atlanta, GA

Master of Science in Computer Science, Concentration in Artificial Intelligence

University of Illinois, Urbana – Champaign

Bachelors of Science in Computer Engineering

WORK EXPERIENCE

Invisible Technologies

NYC, NY

FDE @ Apollo Global Management

2026 – Present

- **Designed and shipped an MCP authorization gateway** that routes AI agents to downstream tool servers (Workday, ServiceNow, HR knowledge, compliance) with Okta JWT validation, tool-scope filtering. Auth elicitation flows, and structured audit logs capturing tool inputs/outputs and user identity.
- **Delivered a production RAG compliance chatbot** (Azure AI Search + GPT-4o, FastAPI, Next.js) from auth and mobile-ready UI. Exposed it to agents via MCP and a LRTC gateway so Claude/Cursor-style clients can query grounded, cited policy answers through the gateway.
- **Authored a shared observability library (corptech-ai-observability)** adopted across 6+ AI services: profile-based instrumentation for MCP servers, gateways, and RAG apps with Datadog LLM Observability, OpenTelemetry tracing, and PII redaction replacing duplicated per-service telemetry.

Northern Trust

Chicago, IL

AI Developer

2023-2026

- Architected AI system using distributed agents, **Python, and LLMs(GPT-4, LangChain, Autogen)** that automated legacy code migration, reducing manual conversion time by 85% across 500K+ lines of code and saving \$1.2M in engineering costs using evaluation metrics, writing custom evals to measure accuracy and reliability, and prompt tuning to improve model outputs by 15%.
- Built enterprise ML-driven classification pipeline for SNOW incident tickets using Azure Machine Learning and Cognitive Search with (RAG) framework automating ticket categorization, reduced triage/resolution time, and enabled natural language Q&A providing support staff with resolution steps and client inquiry responses.
- Orchestrated cloud infrastructure on **Azure using Kubernetes, Terraform, and Docker**, reducing cloud costs while improving system reliability and scalability

INDEPENDENT TECHNICAL PROJECTS

Scientific Paper Retrieval-Augmented Generation System | Python, FAISS, Transformers, Langchain

- Designed a semantic embedding and retrieval pipeline for research papers using transformer models and FAISS vector search.
- Implemented context-aware question answering and text summarization, improving retrieval relevance by 35% over baseline keyword search.
- Integrated dataset evaluation metrics and prompt-tuning loops for response quality analysis.

GenAI-powered Multi-Agent Sales CRM | Firebase, Grok, MCP, Tensorflow

- Built conversational AI agents leveraging LLMs for dialogue generation, intent classification, and personalized outreach.
- Designed language-data experiments to evaluate dialog coherence and user engagement metrics.